

Supporting Information for

Martini 3 Coarse-Grained Force Field for cholesterol

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S1. Supplemental Methods

S1.1 Molecular dynamics simulation analysis.

All simulations were analysed by making use of in-house developed Python 3 programs using the MDAnalysis package¹. We also used the IPython², numpy³, SciPy⁴, scikit-learn⁵, Voro++⁶, and matplotlib⁷ packages for scientific computing in Python. Visualisation and rendering of the simulations were performed with the molecular graphics viewer VMD⁸. Analysis methods part of the GROMACS tool suite were also used. The last 2 μ s of each CG simulation and 1 μ s of each AA simulation were used for analysis unless otherwise stated. Forward mapping of AA simulations was performed with an in-house developed Python 3 program using the MDAnalysis package.

— *Area per lipid (ApL)* was calculated based on Voronoi tessellation of the membrane surface, as described elsewhere⁹. 2D Voronoi tessellation of the membrane surface was carried out, considering periodic boundary conditions, and using the first bead of each acyl chain as the cell centres. For the AA simulations, the centre of geometry of the atoms corresponding to the first CG beads of each acyl chain was used. The Voro++ open-source library⁶ was used to this end, with modified bindings for Python and for the generation of 2D tesserae. The area per lipid was obtained by summing the Voronoi cell areas corresponding to the two lipid acyl chains and then averaging over lipid type and simulation time. System-wide ApL was calculated from the average box area divided by the number of lipids in each leaflet.

— *Octanol–water partition free energies* were calculated from the individual CG solvation free-energies into each solvent, as described elsewhere¹⁰. Interaction parameters and pressure coupling were kept as for normal CG runs. A stochastic dynamics scheme¹¹ was employed at a time step of 20 fs and a friction constant of 1 ps⁻¹, coupling the temperature to 298 K. Pressure was isotopically coupled to 1 bar using the Parrinello-Rahman barostat¹². Single solute molecules were decoupled from solvent boxes of either 898 Martini waters or 431 octanol molecules hydrated with 38 Martini waters. Decoupling was performed in 11 steps of 25 ns each by scaling down solute–solvent Lennard-Jones interactions (Martini water and octanol are not explicitly charged, and therefore do not establish Coulombic interactions with solutes). Partition free-energies and associated error estimates were obtained by the multistate Bennett acceptance ratio (MBAR) method¹³, using the pymbar v3.03 Python package¹⁴.

— *Solvent-accessible surface area (SASA) and Connolly surfaces* were determined using the GROMACS tool gmx sasa. For atomistic references the Van der Waals (VdW) radii from the work of Rowland and Taylor¹⁵ were used, whilst for Martini molecules, the VdW radii for R-, S-, and T-beads are 0.264, 0.230, and 0.191 nm, respectively¹⁶.

— *Membrane z-density profiles* were calculated by determining the target atom's distance to the nearby POPC phosphate layer and histogramming the z-distance component. The nearby POPC phosphate layer was defined by any POPC phosphate within 15 Å of a given target molecule, updated at each timestep (in AA simulations these were taken to be the PC phosphorus atoms, and in CG simulations the PC PO4 beads).

— *Membrane thickness* was determined by calculating the distance between the centres of geometry of each leaflet's phosphate layer. Membrane thickness was defined as the z component of this distance.

— *The order parameter, S*, is a geometric parameter that expresses lipid tail order as a measure of average alignment with the membrane normal. The order parameter used here is the extension to coarse-grain of the concept of lipid C-H bond order parameter and is defined as:

$S = \frac{1}{2} (3 \langle (\cos \theta)^2 \rangle - 1)$, where θ is the angle between the vector along a particular lipid tail bead pair and the normal of the bilayer, which we approximate as the system's z-axis. Lipid tail order parameters were calculated for each lipid tail bond, of each independent lipid as described previously⁹.

— *The cholesterol membrane tilt angle* was calculated with the LiPyphilic toolkit as described elsewhere¹⁷. Briefly, given that the bilayer lies perpendicular to the z-axis, the orientation of cholesterol was defined as the angle between the z-axis and the vector from the *ROH* to the *C1* bead.

— *Spatial distributions of cholesterol neighbours* in the bilayer plane were obtained from lipid bilayer simulations containing cholesterol as described elsewhere¹⁸. Only cholesterol within 1.0 nm of lipid phosphate groups were counted either as distribution references or as neighbours — this selection excludes horizontally tilted cholesterol in the bilayer core from the analysis.

— *Root-mean-square-displacements (RMSD)* were calculated for each cholesterol structure in comparison between reference CG mapped CHARMM36 atomistic simulations and runs with the Martini 3 model here developed. RMSD values were calculated from the ring structures only, overlapped by rotationally and translationally fitting the frame beads. Error estimation was done via bootstrapping of the values obtained for each conformation sampled.

— *The temperature calculation of different lipid groups* was performed as described elsewhere¹⁹. Briefly, the GROMACS *traj* tool was used with .trr trajectory files that contained velocities. The obtained values for molecules that do not contain any constraints can be readily used as *traj* correctly accounts for the molecules' degrees of freedom. For cholesterol, due to the many massless virtual sites and constraints, corrections to the value were done to account for the degrees of freedom. For the model developed in this work, the nine cholesterol CG beads would result in $3N = 27$ degrees of freedom. The 5 massless virtual particles (15 degrees of freedom) and 3 constraints in the topology (3 degrees of freedom) result in the loss of 18 degrees of freedom. To correct for this the value obtained by *traj* was scaled by $27/9$.

— *The cholesterol flip-flop rate* was calculated with the LiPyphilic toolkit as described elsewhere¹⁷.

— *The cholesterol-protein interaction residence time* was calculated with the PyLipID toolkit as described elsewhere²⁰. A dual cutoff of 0.5 and 0.8 nm was used.

S1.2 Virtual site construction.

Virtual site construction followed the steps described in the parameterization of cholesterol for Martini 2 by Melo *et al*¹⁸. Parameters for virtual site construction were obtained from a trajectory of the CHARMM36 cholesterol in a fluid DOPC bilayer. The positions of cholesterol's ring particles were averaged after rotationally and translationally fitting the entire molecule to the *R1-R2-C1* frame, and their placements relative to bead *R1* decomposed as a function of vectors *R1-R2* and *R1-C1*, according to $i = j + a \vec{jk} + b \vec{jl} + c (\vec{jk} \times \vec{jl})$, where a and b are the two scalar coefficients of the combined vectors and c is the coefficient of the added cross product vector (effectively, the out-of-plane contribution). As with previous cholesterol models, nonbonded intramolecular interactions between the ring particles were switched off. The process distributed the masses of the molecule through the frame beads so as to keep the center of mass located as closely as possible to the original atomistic center of mass.

S1.3 Lipid Membrane systems.

Description of the lipid membrane systems tested in this study. AA simulations were run for at least 1 μ s and CG for 10 μ s unless otherwise stated.

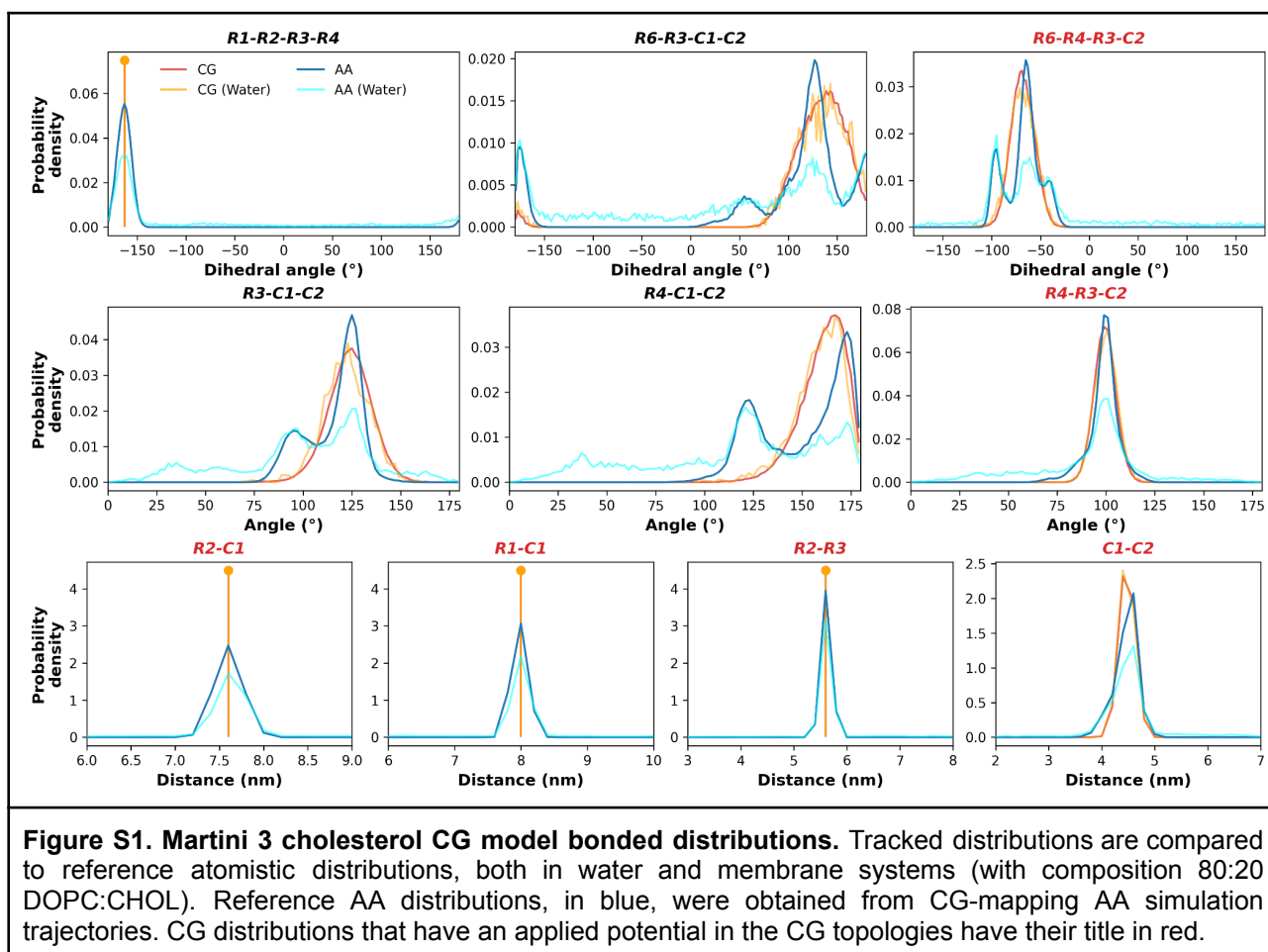
Table S1. Lipid membrane systems tested in this study.

Compositions	Models	Temperature (K)	Size (nm ³)	PBC
85:15 DOPC:CHOL	CHARMM36	300	10 x 10 x 10	Cubic
85:15 DPPC:CHOL	CHARMM36	300	10 x 10 x 10	Cubic
70:30 DPPC:CHOL	CHARMM36	300	10 x 10 x 10	Cubic
85:15 DOPC:CHOL	Martini 2	300	10 x 10 x 10	Cubic
85:15 DPPC:CHOL	Martini 2	300	10 x 10 x 10	Cubic
70:30 DPPC:CHOL	Martini 2	300	10 x 10 x 10	Cubic
85:15 DOPC:CHOL	Martini 3	300	10 x 10 x 10	Cubic
85:15 DPPC:CHOL	Martini 3	300	10 x 10 x 10	Cubic
70:30 DPPC:CHOL	Martini 3	300	10 x 10 x 10	Cubic
85:15 DOPC:CHOL	Martini 3	300	25 x 25 x 10	Hexagonal
85:15 DPPC:CHOL	Martini 3	300	25 x 25 x 10	Hexagonal
70:30 DPPC:CHOL	Martini 3	300	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DLiPC	Martini 3	290	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DOPC	Martini 3	290	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DLiPC	Martini 3	300	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DOPC	Martini 3	300	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DLiPC	Martini 2	290	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DOPC	Martini 2	290	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DLiPC	Martini 2	300	25 x 25 x 10	Hexagonal
42:30:28 DPPC:CHOL:DOPC	Martini 2	300	25 x 25 x 10	Hexagonal
0-50 CHOL:DPPC (10 sims.)	Martini 3	323	10 x 10 x 10	Cubic
0-50 CHOL:POPC (10 sims.)	Martini 3	323	10 x 10 x 10	Cubic
0-50 CHOL:DPPC (10 sims.)	Martini 3 (OpenMM)	323	10 x 10 x 10	Cubic
0-50 CHOL:POPC (10 sims.)	Martini 3 (OpenMM)	323	10 x 10 x 10	Cubic
0-50 CHOL:DPPC (10 sims.)	Martini 2	323	10 x 10 x 10	Cubic
0-50 CHOL:POPC (10 sims.)	Martini 2	323	10 x 10 x 10	Cubic

S2. Description of Videos

Video S1. Top view of Martini 3 70:30 DPPC:CHOL, 85:15 DPPC:CHOL and 85:15 DOPC:CHOL lipid bilayer simulations at 300 K. DPPC first acyl-chain beads (*C1A*, *C1B*) are represented in grey. Cholesterol hydroxyl bead (*ROH*) is represented in red. Note the transient formation of DPPC gel domains (visible through the hexagonal acyl-chain arrangement) over the course of the simulation. The last 500 ns are shown.

S3. Martini 3 cholesterol bonded parameters



S3.1. Complete set of Martini 3 cholesterol CG parameters.

In this section, the entire set of Martini parameters is laid out. The bead types dictate each particle's nonbonded Lennard-Jones interactions, as per the Martini 3 interaction table¹⁶. None of the considered particles carries any charge. Bond stretching potentials are harmonic and angle bending potentials are cosine harmonic. Non-periodic (improper) dihedral bending potentials are harmonic. Intramolecular first-bonded neighbours do not interact via nonbonded interactions. All particles in ring moieties (all the S-type particles) are also excluded from intramolecular nonbonded interactions with one another. A mass of zero is assigned to virtual-site particles. Virtual site construction parameters are given as a, b, and c coefficients. Parameters are available in GROMACS-compatible format at

<https://github.com/Martini-Force-Field-Initiative/M3-Cholesterol-Parameters>.

Table S2. Bead numbering, name, type and mass for the cholesterol model developed in this study.

Number	Name	Type	Mass (a.m.u.)
1	<i>ROH</i>	P1	0.0
2	<i>R1</i>	SC4	159.8
3	<i>R2</i>	SC3	157.0
4	<i>R3</i>	SC3	0.0
5	<i>R4</i>	SC3	0.0
6	<i>R5</i>	TC2	0.0
7	<i>R6</i>	TC2	0.0
8	<i>C1</i>	C2	115.2
9	<i>C2</i>	C2	72.0

Table S3. Bonded and constraint parameters for the cholesterol model developed in this study.

Bond	Equilibrium length (nm)	Force constant (kJ/(mol nm ²))
<i>C1 – C2</i>	0.440	15000
Linear constraint	Length (nm)	
<i>R1 – R2</i>	0.34797	
<i>R1 – C1</i>	0.78504	
<i>R2 – C1</i>	0.75012	
Harmonic angle potential	Equilibrium amplitude (°)	Force constant (kJ/mol)
<i>R4 – R3 – C2</i>	99.0	250
Improper dihedral potential	Equilibrium amplitude (°)	Force constant (kJ/(mol rad ²))
<i>R6 – R4 – R3 – C2</i>	-70.0	50

Table S4. Martini 3 Cholesterol CHARMM36 mapping key. Some beads are mapped as linear combinations of atomic positions.

Number	CG Bead Name	CHARMM36 atom name
1	<i>ROH</i>	C2, C2, C4, C4, O3, O3, O3, O3
2	<i>R1</i>	C6, C6, C6, C6, C6, C5, C7, H6, H7A, H7B
3	<i>R2</i>	C9, H9, C10, C1, H1B, H1A, C9, H9, C10, C1, H1B, H1A, H1B, H1A, H1A, H1B
4	<i>R3</i>	C15, C15, C15, C15, C15, C14, C14, C16, C16
5	<i>R4</i>	C11, C12
6	<i>R5</i>	C19
7	<i>R6</i>	C18
8	<i>C1</i>	C20, H22A, H22B, H20, H21A, H21B, H21C
9	<i>C2</i>	C23, C24, C25, C26, C27

S4. Supplemental results

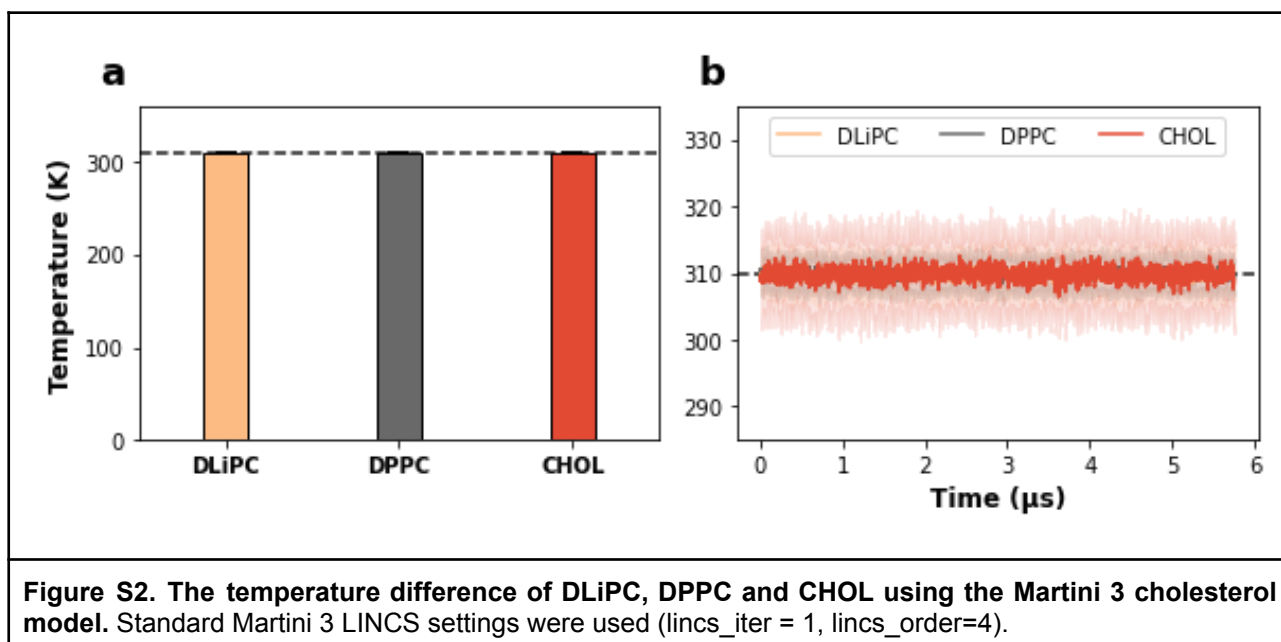


Figure S2. The temperature difference of DLiPC, DPPC and CHOL using the Martini 3 cholesterol model. Standard Martini 3 LINCS settings were used (lincs_iter = 1, lincs_order=4).

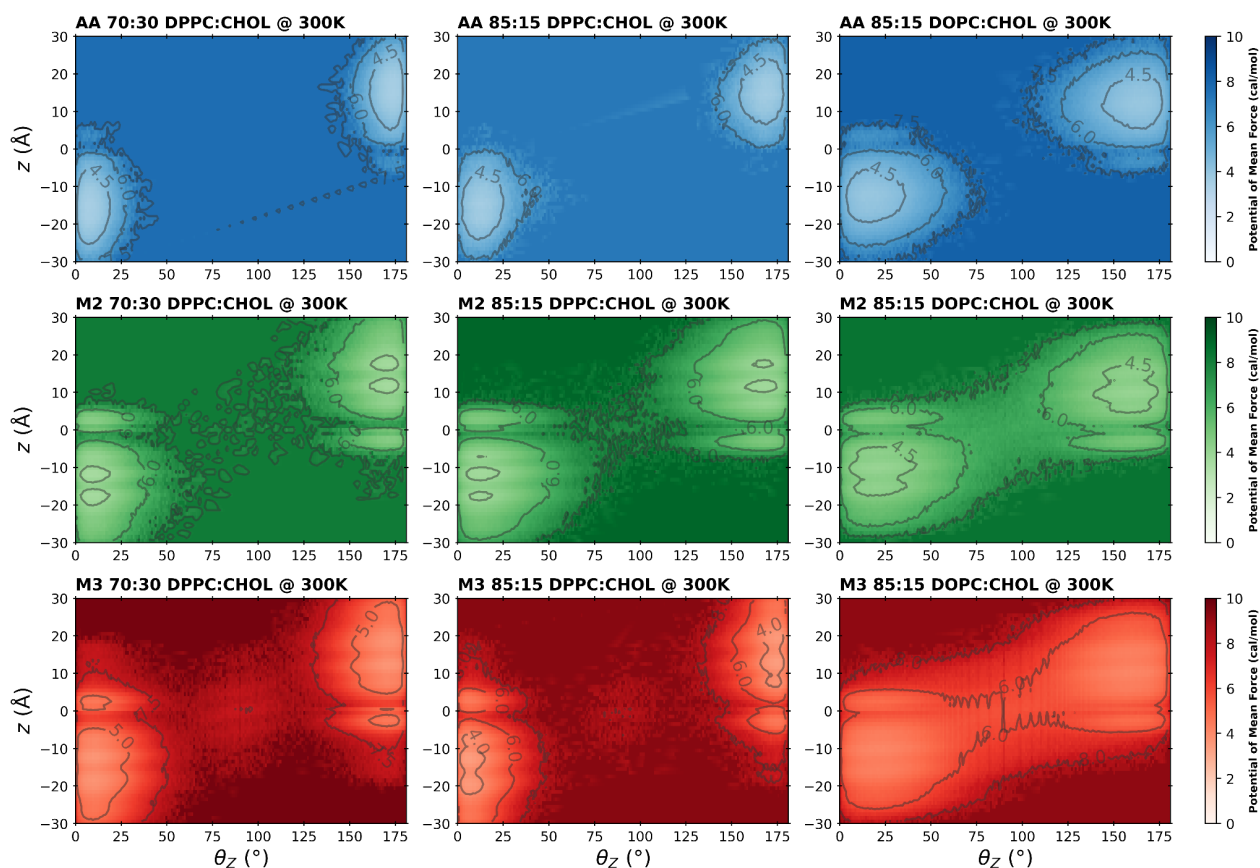


Figure S3. Cholesterol height and orientation potential of mean force (PMF). Cholesterol height and orientation PMFs calculated for our AA reference (blue), Martini 2 (green) and the Martini 3 model (red).

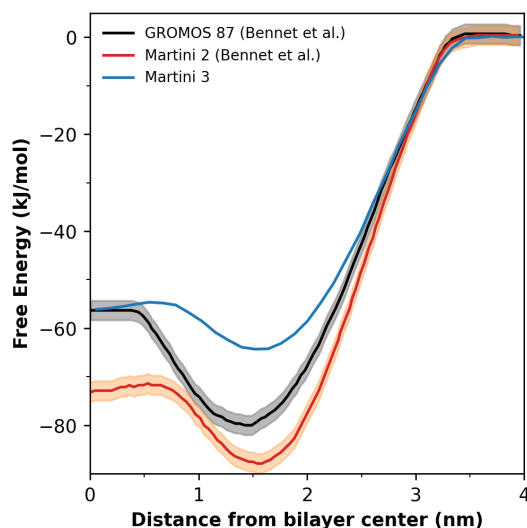


Figure S4. CG PMF of pulling cholesterol (*ROH* bead) from bulk water to the centre of a POPC bilayer. PMF was set to zero in bulk water. Error bars are the standard error from the mean of the three PMF replicas. The PMFs for Martini 2 and GROMOS 87 (GROMOS 87 cholesterol with Berger et al. force-field for phospholipids) were obtained from Bennet et al.²¹. Error bar is smaller than the curve line for the Martini 3 PMF.

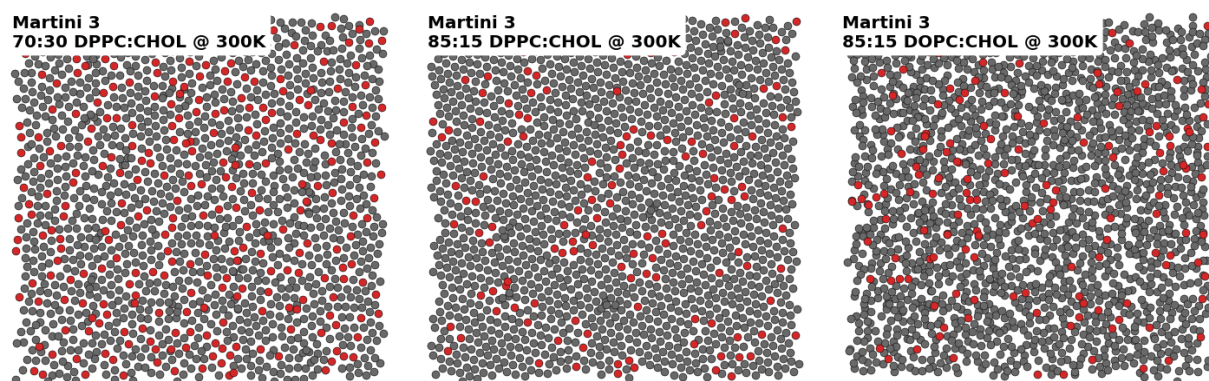


Figure S5. Top view snapshots of lipid bilayer simulations containing cholesterol comparing the gel-phase formation behaviour. PC first acyl-chain beads (*C1A*, *C1B*) are represented in grey. Cholesterol hydroxyl bead (*ROH*) is represented in red. Note the formation of gel domains (visible through the hexagonal acyl-chain display).

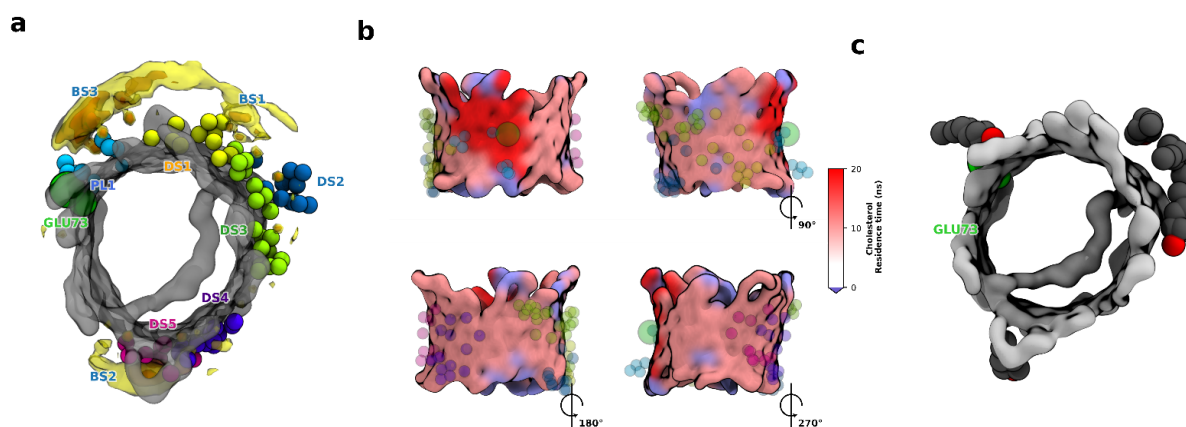


Figure S6. Cholesterol recognition and binding by VDAC1. (a) Occupancy map of Martini 3 cholesterol interactions with VDAC1. (b) VDAC1 contact surface coloured by cholesterol residence time. (c) Snapshot of Martini 3 cholesterol bound to several detected VDAC1 binding sites.

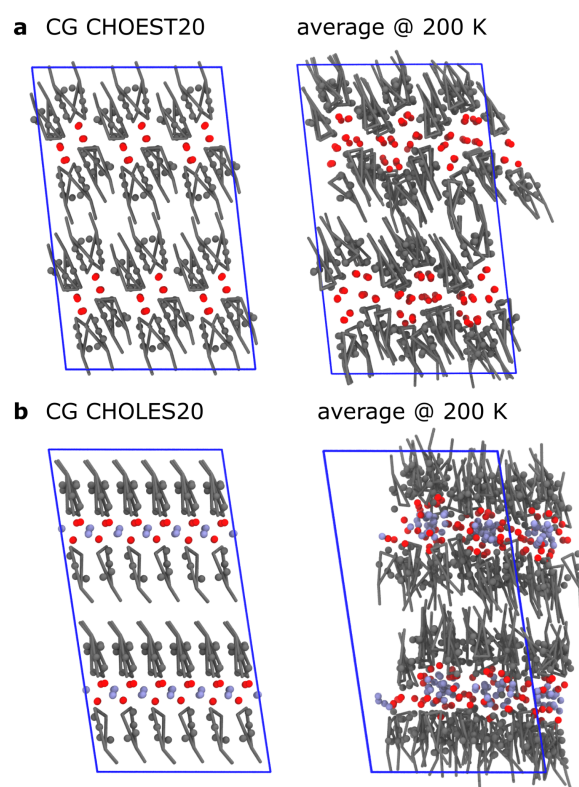


Figure S7. CG cholesterol crystal structure simulations. Supercell of (a) pure cholesterol (CHOEST20) and (b) cholesterol monohydrate at the initial CG mapping of the crystal structures (left) and averaged over 200 ns of CG simulation at 200 K (right). The tiny water bead in cholesterol monohydrate is depicted in light blue.

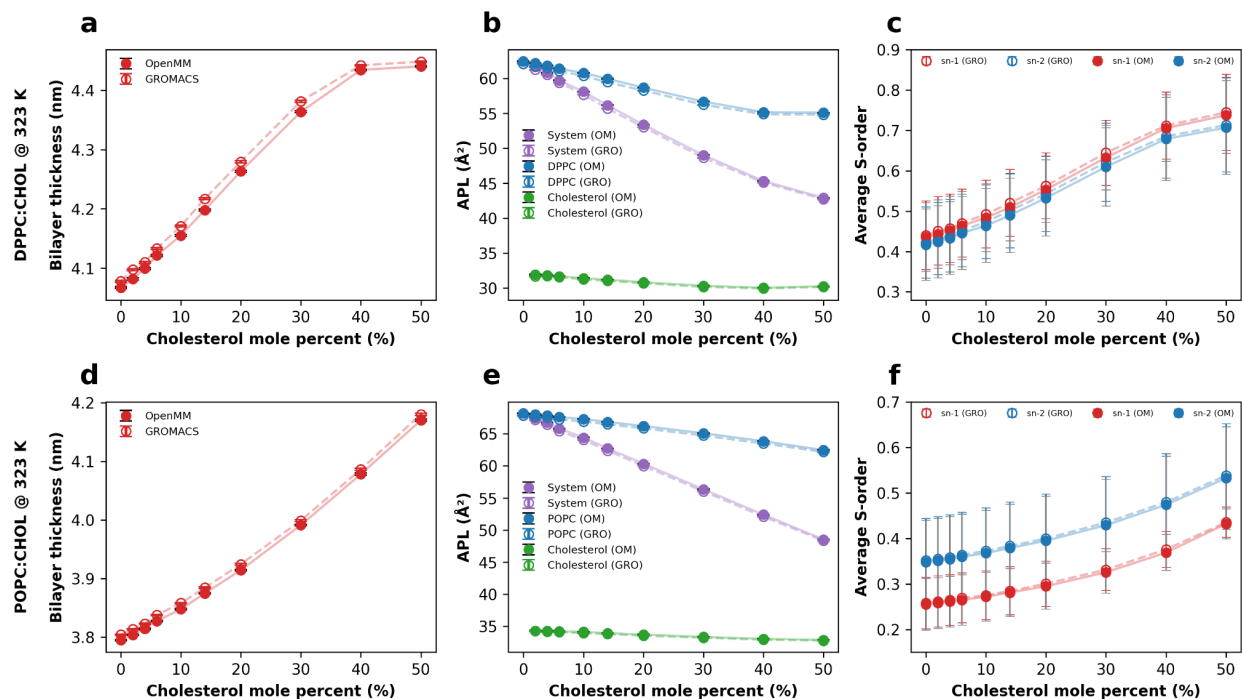


Figure S8. Comparison between OpenMM and GROMACS Implementations of the Martini 3 Cholesterol model. Effect of increasing concentrations of cholesterol on (a,d) bilayer thickness, (b,e) APL and (c,f) average acyl chain order of DPPC (top row, a,b and c) and POPC (bottom row, d,e and f) membrane bilayers at 323 K. Error bars represent the standard deviation estimated by bootstrapping

S5. Supplemental References

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